

Aeration and Foam Control in Baker's Yeast Production: Mapping Patents

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Abstract: A key ingredient in the baking industry, baker's yeast must be produced under strict controlled conditions. High yields of baker's yeast cannot be attained unless a huge quantity of air is injected into fermentation vats. This review of 245 patent specifications shows that inventors have paid much attention to the distribution of fine air bubbles in order to optimize oxygen transfer to the yeast cells. Technical solutions to reduce energy costs associated with aeration are also proposed. Intense aeration caused foaming problems, so mechanical destruction of foam was first proposed until inventions on specific chemical antifoams were patented. In recent years, the development of cheaper and more efficient foam control techniques has remained an issue. Aeration during yeast growth in tanks impairs its fermentative activity in anaerobic bread dough. Since the beginning of the 20th century, massive adoption of air-grown fed-batch baker's yeast probably encouraged sugar addition to stimulate yeast gassing activity in pan bread characterized by high loaf volume, especially those prepared under short dough fermentation conditions.

Keywords: aeration, baker's yeast, bread, foam control, patents

Introduction

Baker's yeast is a concentrate of living microorganisms that is used as a starter in bread dough fermentation. This typical baking ingredient was formerly a by-product of alcohol and beer manufacturing and, over the years, its manufacturing process has changed considerably (Gélinas 2010). In the end of the 19th century, growth media began to be aerated and the production of baker's yeast departed from anaerobic alcohol manufacturing. Based on the weight of the raw materials, aeration is generally considered as the most important single factor to increase yeast yield that is from 12% to 35% with a nonoptimized aeration process (Hart 1914). In modern yeast production, yeast biomass yield is about 100% under controlled aeration and fed-batch conditions (Suomalainen 1963; Hoogerheide 1969).

In the course of yeast cell growth, aeration is so intense that oxygen ranks first among ingredients in the fermentation tanks. Over the years, the latter have been carefully designed so oxygen and liquid medium remain in contact for the longest time possible (de Becze and Liebmann 1944). However, oxygen limitation is likely to occur more or less rapidly depending on the medium composition and the height of the reactor (Sweere and others 1988). The demand of yeast cells for oxygen is so great that they are likely to grow under some oxygen-deficit conditions (Gélinas 1988; Gélinas and others 1989). Blanco and others (2008) have addressed critical environmental factors that may result in lack of dissolved oxygen in fermentation tanks: low or insufficient

aeration, high sugar feeding, high baker's yeast concentration in the fermentation broth, and high temperature due to lower oxygen solubility.

In the early years after adoption of aeration, the baker's yeast trade was concerned by huge energy costs associated with aeration of fermentation vats and mainly technical solutions were described in the patent literature (de Becze and Liebmann 1944; Irvin 1954; Pyke 1957). It has been estimated that aeration contributes to 10% to 20% of the total cost of baker's yeast (de Becze and Liebmann 1944). Technical information on oxygen transfer in submerged fermentation has been addressed by Hixson and Gaden (1950) and Maxon and Johnson (1953). These papers were later updated, for example, by Garcia-Ochoa and Gomez (2009), who stressed that continuous supply of oxygen in bioreactors was needed because of its low solubility in aqueous solutions. Intense aeration also causes major foaming problems in the fermentation vats so this must be controlled with antifoam devices. Technology related to foam destruction in fermentation processes has been compiled by Pyke (1957), Bryant (1970), Evans and Hall (1971), Hall and others (1973), Viesturs and others (1982), Ghildyal and others (1988), and Vardar-Sukan (1998). Mechanisms of foam destruction have been reviewed by Denkov (2004) and Höfer and others (2012).

Nowadays, baker's yeast production may be considered as a ripened technology but technical problems associated with aeration and foam control in baker's yeast production do not appear to have been completely solved. Much is left to explore, especially to reduce costs and improve equipment. As of late, no comprehensive review has been published on aeration and foam control, although the latter are key processes in baker's yeast production.

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Inventions are constantly being claimed and are timely answers to technical problems in the food industry. Studying the evolution of patenting activity is an original way to better understand short- and long-term trends and technical problems (Gélinas 2010). This gray literature is rarely considered in reviews of the scientific and technical literature. The aim of this review was to follow the interest of inventors for aeration and foam control as part of baker's yeast production. It was hypothesized that, in retrospect, intense aeration in fermentation tanks during baker's yeast growth had major effects on bread quality.

Patents

Background information and search criteria

A privilege granted for about 20 y, a patent is a legal document that fully describes innovative solutions to technical problems. Identical patent specifications may be filed in various countries, but some conditions do apply. In early patents granted in some countries, commercial applications were highly praised, although novelty aspects were not always considered. This review is based on granted patents, although 2 nonaccepted specifications were found and are cited: GB 191007640 and GB 472622. In the patent literature, patent specifications describing the inventions are numbered according to country, as Great Britain in the above examples.

This review covers aeration and foam control disclosed in the patent literature and filed between 1900 and 2014. Reference to earlier patents was adapted from a companion study (Gélinas 2010). Patent specifications are reported here according to the filing date or priority date, not the publication date of the granted patent. Efforts have been made to spot and combine similar patent applications filed in different countries and describing variations of the same basic invention. These are called “patent families.” Sometimes the same invention was described in patents filed in different countries and even granted to different applicants or inventors without reference to the priority file. Careful reading and comparison of patent specifications was essential to get a true picture of patenting activity over the years (Gélinas 2012). Such a large puzzle included about 560 individual patent specifications, mainly written in German, English, French, Danish, and, to a lesser extent, in Swedish, Dutch, and Finnish. Most documents were retrieved from search data bases, especially esp@cenet and Depatis, although some early patents were traced according to Wagner (1936).

During the preparation of this review, efforts were made to retrieve only inventions relevant to baker's yeast production, although many patents have been published on aeration and foam control in miscellaneous other aerobic fermentation media. Molasses-based media for baker's yeast production are completely different from media used in the production of antibiotics with a *Penicillium* mycelium, for example. Oxygen diffusion is lower in such highly viscous fermentation media, so the design of aeration units is different (Müller 1976).

More details on the patent literature are available in companion papers covering specific aspects of the patent literature on baker's yeast production: early inventions from the 19th century (Gélinas 2010), growth media (Gélinas 2012), and fermentation control (Gélinas 2014).

Evolution of Patenting Activity According to Year Number of patents

Overall, 245 patent specifications describing unique inventions were found on aeration and foam control in baker's yeast production, including 13 filed prior to 1900 (Table 1). Most of them, or

148, concern aeration, particularly the addition and distribution of finely divided air bubbles in fermentation vats. In addition, 97 patent specifications were filed on foam control, either by mechanical or chemical ways. Although 1890 to 1900 may be considered as the true beginning of intense aeration for growing baker's yeast, inventors showed the most interest for these matters between 1930 and 1934 (Table 1; Figure 1). As mentioned earlier, data shown in Table 1 and Figure 1 are based on patent families, which combined later additions to the main patented invention. For aeration and foam control, 60 complementary or additional patents associated with main inventions were found, which is comparable to growth media (69 patents) and fermentation control (61 patents).

Interest in foam control was less than for aeration. It also developed later, around 1900, which indicates that intense aeration was practiced during that period (Figure 1). Prior to 1950, mechanical systems were popular solutions to control foam (Table 1). This was superseded by chemical antifoams so, until the end of the 1990s, highly specific chemical formulations have regularly been patent-protected for baker's yeast production.

Figure 2 shows trends in patenting activity on aeration and foam control, compared to media and fermentation control: inventions on yeast growth conditions peaked from 1920 to 1934. Interest for aeration and foam control remained high between 1910 and the end of the 20th century, lasting longer after the 1930 to 1934 peak compared to growth media or fermentation control.

Inventors

Country of origin

Since the 1870s, inventors of German origin were the most active with 120 patent applications or half of patented inventions in this field (Table 2). To a lesser extent, citizens from Austria (37 patents) and the United States (32 patents) also showed interest in aeration and foam control. Citizens from 9 other countries also contributed to the peak of 41 patent applications filed between 1930 and 1934 on this subject.

Major inventors

In the years 1877 to 1879, Eusebius Bruun from Denmark developed a process where plenty of air was introduced in fermentation vats (Waller 1913). However, according to this author, Bruun never sought patent protection for this invention because he wanted to avoid its full description in a publication. In May 1879, Henry zum Felde appears to have been the 1st inventor to claim patent protection on a specific aeration unit to promote the growth of yeast cells according to Pasteur's teachings (Gélinas 2010).

During the modern era of yeast production, 2 major inventors on aeration merit special attention. Between 1927 and 1960, Eugen Georg Stich filed many patent specifications on innovative solutions to improve aeration of liquid media for yeast production; overall, he is listed as the inventor in at least 70 individual patents issued in several countries, combined here into 13 patent families. For a few patents, this German inventor from Mannheim was linked to a German manufacturer, Norddeutsche Hefeindustrie AG, based in Berlin thus very far from his hometown. Stich also claimed 2 patents on mechanical foam control, described in this review, as well as 1 patent on molasses treatment, 1 on growth media (adjuncts), 1 on seeding conditions, and 2 on fermentation control with some link to aeration (Gélinas 2012, 2014).

Another very active inventor on yeast aeration aspects was Wilhelm Vogelbusch (1887 to 1979) from Vienna, Austria. From

Table 1—Categories of patents on aeration and foam control in baker's yeast production according to filing date.^a

Year	Air			Antifoam			Total
	Control	Circulation	Distribution	Mechanical	Chemical	Distribution	
1875 to 1879	1		1				2
1880 to 1884							
1885 to 1889	1	1					2
1890 to 1894		6					6
1895 to 1899		1	2				3
1900 to 1904			1	1			2
1905 to 1909	1			1			2
1910 to 1914	1	1	4	3		1	10
1915 to 1919	1		2	8	1	1	13
1920 to 1924			2	4			6
1925 to 1929	3	2	9	1		2	17
1930 to 1934	2	6	26	1	2	4	41
1935 to 1939	3	3	6	4		2	18
1940 to 1944		5	1	4	4	1	15
1945 to 1949		1	4	3	2	1	11
1950 to 1954	1		8		5		14
1955 to 1959	1		8		3	1	13
1960 to 1964	2	1	4		1		9
1965 to 1969		3	1	2		1	7
1970 to 1974	4		4	1	2		11
1975 to 1979		1	2	1	3		7
1980 to 1984				1	2		3
1985 to 1989	1	1	2		4		8
1990 to 1994					9		9
1995 to 1999			3		6		9
2000 to 2004		2			1		3
2005 to 2009			1		1		2
2010 to 2014 ^b			1			1	2
Subtotal (before 1900)	2	8	3				13
Subtotal (1900 to 2014)	20	26	89	35	46	16	232
Total	22	36	90	35	46	16	245

^aPatents prepared by the same inventor and representing variations or improvements of the same invention were combined into a single patent for invention. Information on patents filed before 1900 was adapted from Gélinas (2010). Since the publication of the 2010 review, 2 patents by German inventors were found on air circulation: Fernbach (FR 207801, granted on August 23, 1890); Genge (FR 207945, granted on August 30, 1890).

^bProbably underestimated because some patent applications may still be under examination.

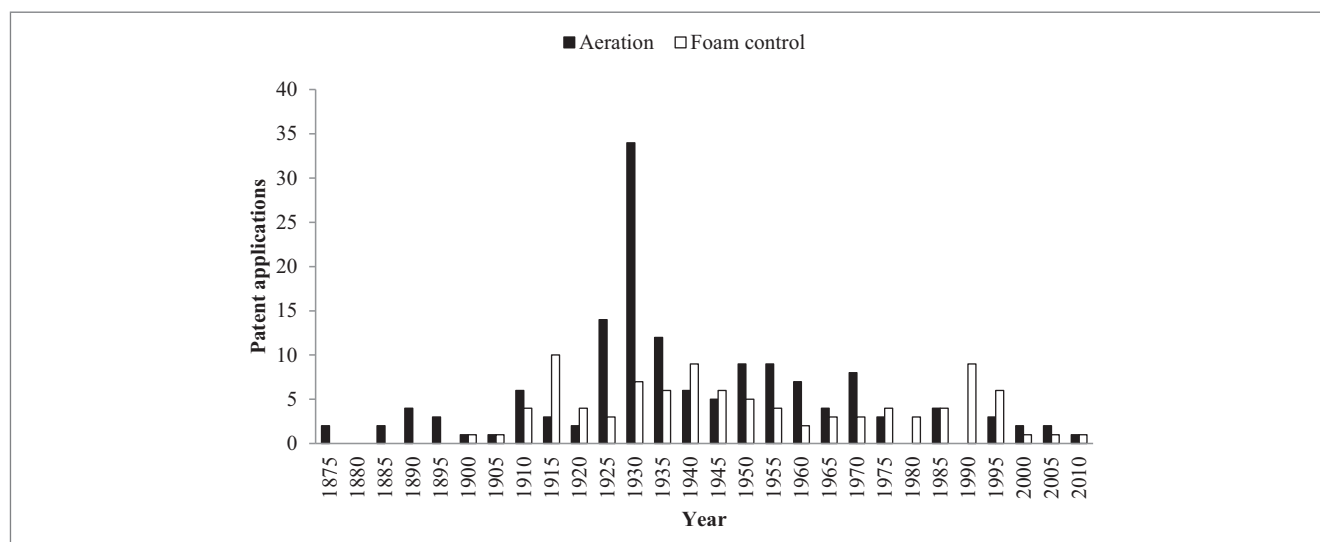


Figure 1—Number of patent applications on aeration and foam control during the production of baker's yeast according to filing date (5 y range). Data for 2010 to 2014 are probably underestimated because some patent applications may still be under examination.

1922 and prior to selling his company to Mautner Markof Presshefefabrik in 1956, he personally claimed not less than 12 patented inventions, or patent families, including 9 on aeration and 3 on foam control. Overall, Vogelbusch was listed as inventor on 32 individual patents, including additional or complementary patents. After 1956, Patentauswertung Vogelbusch GmbH was granted 14 more patents on these matters, notwithstanding many variations described in complementary files.

Companies

According to Table 3, 13 industrial assignees, including the Secretary of Agriculture of the United States, filed at least 3 patents on aeration and foam control, a number similar to other aspects of baker's yeast manufacturing such as growth media (Gélinas 2012) and fermentation control (Gélinas 2014). Among the most important industrial assignees in baker's yeast manufacturing, a major difference was the absence of the Fleischmann/Standard Brands

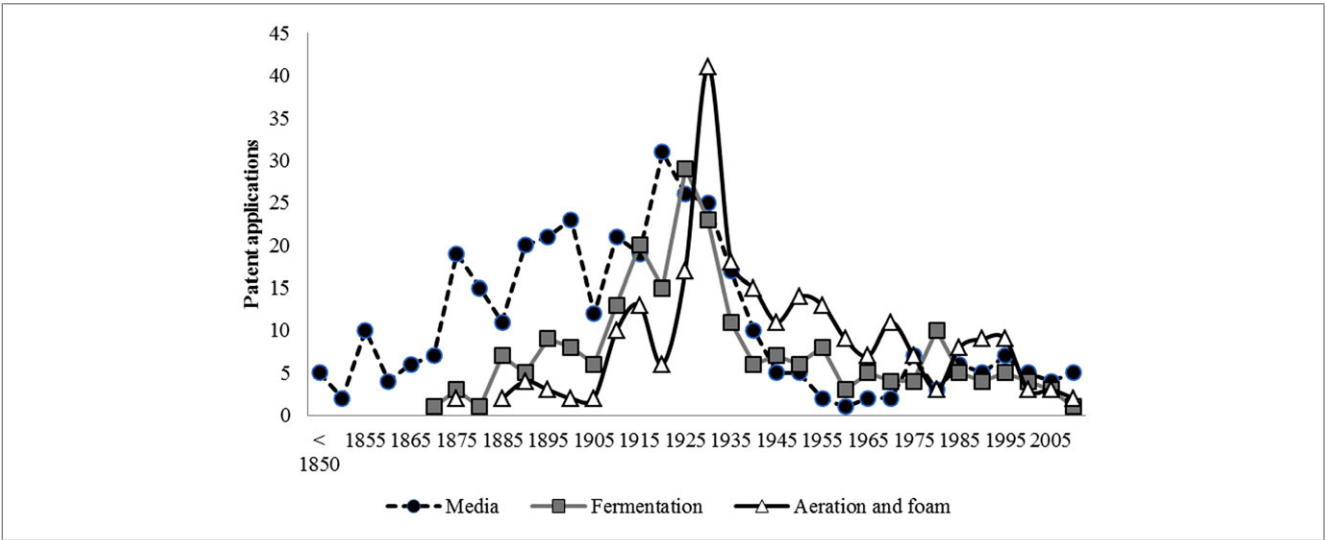


Figure 2—Number of patent applications on media, fermentation control, and aeration and foam control during the production of baker's yeast according to filing date (5 y range). Data on media and fermentation control were obtained from Gélinas (2012, 2014). Since the publication of the latter reviews, the following 8 patents were found and counted above, specifically on growth media (YU 57002 filed in 2002; KR 100522264 filed in 2005; US 20120128853 filed in 2010; RU 2010152223 filed in 2010; CN 102311927 filed in 2011; RU 2012138881 filed in 2012; CN 104087521 filed in 2014), and fermentation control (RO 127019 filed in 2012). Data for 2010 to 2014 are probably underestimated because some patent applications may still be under examination.

Table 2—Country of origin of inventors on aeration and foam control in baker's yeast production according to filing date of patents.

Year	DE	AT	US	GB	SE	NL	FR	CH	AU	CS	DK	JP	Others	Total
Before 1900 ^a	8	1	1				1	1					FI	13
1900 to 1904	1							1						2
1905 to 1909	2													2
1910 to 1914	7	3												10
1915 to 1919	7	3	1			1				1				13
1920 to 1924	4	1		1										6
1925 to 1929	11	1	1	1	1	2								17
1930 to 1934	13	7	2	4	3	2	2			2	2		HU; NO; PL (2)	41
1935 to 1939	9	2	3	1	3									18
1940 to 1944	10	1	3								1			15
1945 to 1949	1	1	8				1							11
1950 to 1954	3	2	7	2			1							14
1955 to 1959	6	5	1				1							13
1960 to 1964	6	2			1									9
1965 to 1969	1	2	1	1				1	1					7
1970 to 1974	4	4						1				2		11
1975 to 1979	3	2										1	DD	7
1980 to 1984	3													3
1985 to 1989	7								1					8
1990 to 1994	8									1				9
1995 to 1999	6		1						2					9
2000 to 2004			2			1								3
2005 to 2009			1										RU	2
2010 to 2014 ^b				1									CN	2
Total	120	37	32	11	8	6	5	4	4	4	3	3	8	245

^aThese data were partly adapted from Gélinas (2010).
Based on Recommended Standards of the World Intellectual Property, the following 2-letter codes for the representation of States have been used for patent numbers and country of origin of inventors: AT = Austria; AU = Australia; CH = Switzerland; CN = China; CS = Czechoslovakia; DD = German Democratic Republic; DE = Germany; DK = Denmark; FI = Finland; FR = France; GB = Great Britain; HU = Hungary; JP = Japan; NL = Netherlands; NO = Norway; PL = Poland; RU = Russia (including Soviet Union); SE = Sweden; US = United States. For patents with inventors from different countries, only the country of the 1st inventor was considered.
^bData for 2010 to 2014 are probably underestimated because some patent applications may still be under examination.

group with only 1 patent on foam control. Overall, 68 companies or industrial groups filed 139 patents on this subject or more than half of total patents, which reveals its commercial importance.

Most of the industrial companies interested in filing patent specifications on aeration or foam control were based in Germany (Table 3). With 22 patents filed between 1971 and 1997 on chemical antifoams, the most active company in foam control during baker's yeast production was Henkel & Cie GmbH, Düsseldorf, Germany. As mentioned above, Vogelbusch GmbH filed 14 patents, mainly in Austria and also in several other European countries nearby.

Licenses

Different applicants sometimes filed identical versions of the same invention in different countries. Such practice was an indication that exclusive rights over patented inventions were licensed to others, generally companies (Gélinas 2012). In some cases, ownership of intellectual rights was difficult to determine and there were true possibilities of overestimating patenting activity.

Especially on aeration, 13 patented inventions were licensed to 6 manufacturers or industrial groups, especially between 1916 and 1945 (Table 4). That number represents about 10% of total patents

Table 3—List of the major industrial assignees on aeration and foam control in baker's yeast production for patents filed between 1900 and 2014.

Assignee (city, country)	Application (filing y)
Fried. Krupp GmbH (Essen, Germany)	Air circulation (1988); air diffusion (1988); antifoam, mechanical (1980)
Hefe-Patent GmbH (Berlin, Germany)	Aeration control (1939; 1959); antifoam, mechanical (1936; 1941)
Henkel & Cie GmbH (Düsseldorf, Germany)	Antifoam, chemical (1971; 1972; 1975; 1975; 1975; 1984; 1985; 1986; 1988; 1989; 1990; 1990; 1990; 1991; 1993; 1993; 1994; 1994; 1995; 1995; 1996; 1997)
I. G. Farbenindustrie AG (Frankfurt, Germany)	Air circulation (1928); antifoam, chemical (1931; 1941)
Norddeutsche Hefeindustrie AG (Berlin, Germany); later (1933) part of Wirtschaftliche Vereinigung der Deutschen Hefeindustrie	Aeration control (1928); air distribution (1929; 1932); antifoam, mechanical (1926)
Phrix-Werke AG (Hamburg, Germany)	Air circulation (1942); air distribution (1944; 1953)
Secretary of Agriculture, the United States	Air circulation (1947); antifoam, mechanical (1947; 1949)
Sinner AG (Karlsruhe-Grünwinkel, Germany)	Air distribution (1931; 1938; 1951)
Strauch & Schmidt (Neisse, Germany)	Air distribution (1925; 1930; 1930); antifoam, mechanical (1918; 1919; 1921)
Svenska Jästfabriks Aktiebolaget (Stockholm, Sweden)	Aeration control (1938; 1963); air circulation (1930); air distribution (1933); antifoam distribution (1937)
Swift & Co. (Chicago, Ill., U.S.A.)	Antifoam, chemical (1950; 1952; 1956)
The Distillers Co. Ltd. (Edinburgh, Scotland, Great Britain)	Air distribution (1932; 1935); antifoam distribution (1930; 1932; 1932; 1968)
Vogelbusch GmbH (Vienna, Austria); Technisches Büro Vogelbusch; Patentausswertung Vogelbusch GmbH	Air circulation (1968; 1975); air distribution (1929; 1955; 1956; 1957; 1958; 1964; 1972; 1972; 1973; 1974); antifoam, mechanical (1969); antifoam distribution (1960)

^a For the sake of uniformity throughout this review, some terms associated with the names of some German or Austrian companies were abbreviated as follows: GmbH (Gesellschaft mit beschränkter Haftung; company with limited liability), AG (Aktiengesellschaft; corporation limited by share ownership), and KGaA (Kommanditgesellschaft auf Aktien; commercial partnership limited by shares).

filed during that period, which is similar to growth media but less than for fermentation control with about 15% of patents leading to licenses (Gélinas 2012, 2014). In 2 cases, no proper reference was made to the priority file or the 1st applicant. In 3 other cases, licenses were filed more than 1 y after the 1st filing date, which is excessive according to patent offices (Table 4).

The number of licenses peaked between 1927 and 1935, which is another indication of the great interest by yeast manufacturers for this technical problem during that period. In many cases, the rights of those inventions were owned by German citizens and licensees were often based in the United States. Hence, most of the licenses, 6 in total, were obtained by the Fleischmann/Standard Brands group. Including the latter licenses and patents, and those reported by Gélinas (2014) for growth media and fermentation control, this company had the same number of licenses and original patents, or 28, on various aspects of yeast growth in fermentation tanks.

Aeration

Optimization of aeration conditions of liquid growth media is a very important matter in baker's yeast manufacturing. Three types of patented inventions were considered: aeration control, air circulation, and air distribution.

Air control

Contrary to other aspects of aeration, little interest was given to the control of air injection into the fermentation tank. Since the early days, it was known that strong aeration improved yeast yield. Priority was given on the best techniques to feed yeast with as much oxygen as possible. However, cost associated with the use of large quantities of air was also a major concern. Table 5 describes 20 patents issued between 1900 and 2014 on aeration control.

Stich and Rosenqvist both proposed to improve control of air delivery. Ozone-enriched air was proposed in 1939, although this had already been suggested by Rainer in 1879 (Gélinas 2010). Worth mentioning is a patent specification filed in 1928 by Norddeutsche Hefeindustrie AG which recommends to increase the aeration rate during the course of the production to compensate for higher oxygen demand resulting from yeast cell multiplication.

In an interesting patent-protected innovation by Braasch and Braasch, air was enriched with carbon dioxide to improve yields. In later years, Chen and Gutmanis (1976) showed that carbon dioxide in fermentation tanks did not inhibit yeast growth. Some inventors proposed recirculating exit gas containing residual oxygen and carbon dioxide excreted by the growing yeast. In other cases, liquid medium itself was recirculated with the exhaust gas. In 2 patents, filed in 1959 and 1963, continuous measurement of dissolved oxygen was proposed to control aeration online in combination with optimal sugar feeding rate.

Air circulation

Inventors considered that a large proportion of injected air was not assimilated by yeast cells and lost to the atmosphere. Table 6 presents 35 innovative solutions on air circulation devices. Half of them were filed during the golden years of the development of modern baker's yeast manufacturing, between 1920 and 1944. To optimize circulation of air through a column of watery growth medium, compressed air was generally introduced at the bottom of the fermentation tank, although some inventions proposed to supply air from the top. Liquid was sometimes pumped and moved to an aeration unit placed outside of the main vat. Such aeration units were divided into several compartments or equipped with specific deflectors that created violent agitation of the liquid medium, to increase oxygen transfer. In general, a single source of air injection was proposed but some inventors recommended multiple entries, sometimes through a series of connected fermentation vats. The aeration system was also used to provide nutrients to the growth medium. Foam was also degassed and recirculated through the bottom of the vat. Several air circulation systems were specifically developed to reduce foaming problems.

More patents on air circulation have been issued in addition to those described in Table 6. For example, specific fermentation vessels for the continuous addition-withdrawal process were described by Stich (1951) and Olsen (1960); these patents are addressed in Gélinas (2014). Especially after 1950, many fermentation tanks were patent-protected for miscellaneous fermentations, not specifically for baker's yeast. The latter are not considered in this review, for example, inventions described by Jonáš and Kořán

Table 4—Cases of licenses (1900 to 2014) in which several patent applicants claimed the same invention on aeration and foam control in baker's yeast production.^a

Priority date	Inventor or 1st to file; patent nr.	Licensee; patent nr.; filing date	Link with the priority file	Subject
August 20, 1916 (DE)	Hayduck; Verein der Spiritus-Fabrikanten in Deutschland; DE 303801	The Fleischmann Co.; US 1449114; CA 233379; first filed November 1, 1920	Late filing date, probably extended due to the war	Antifoam, mechanical
January 28, 1927 (DE)	Stich; DE 567518	Norddeutsche Hefeindustrie AG; DE 594361; DE 594192; DE 594193; DE 594194; DE 594195; first filed October 25, 1927	Additions to the 1st file	Air distribution
October 25, 1927 (DE)	Stich; Norddeutsche Hefeindustrie AG; DE 594361; FR 680303 (DK 47748; US 1792450; GB 340644)	AG Ignaz Kuffner & Jacob Kuffner für Brauerei, Spiritus- und Presshefefabrikation Ottakring-Döbling, AG für Spiritus- und Presshefe-Industrie Wolfrum, M. Fischl's Söhne und Vereinigte Mautner Markhof'sche Presshefe Fabriken; AT 134100; filed August 5, 1929	No proper reference to Stich; the 2nd patent was licensed	Air distribution
February 20, 1930, and December 6, 1933 (NO)	Jansen; AT 138613 (DK 47315; DK 47373; GB 366753; FI 16624); US 2056576 (FR 782281)	Norddeutsche Hefeindustrie AG; DE 621566; DE 666321; filed May 4, 1930, and December 5, 1936	No proper reference to Jansen	Air circulation
January 7, 1932 (GB)	Lockey; The Distillers Co. Ltd.; GB 387486	Standard Brands Inc.; US 2104330; US 2176558; filed January 6, 1933		Air distribution
January 10, 1932 (DE)	Braasch and Braasch; DE 605912 (GB 395610; FR 748594)	Standard Brands Inc.; US 2092151; filed February 18, 1936	Late filing date	Air distribution
January 16, 1932 (DE)	Braasch and Braasch; DE 655729	Standard Brands Inc.; US 2013456; CA 361390; filed January 14, 1933 (US)		Aeration control
June 10, 1932 (GB)	Everett; The Distillers Co. Ltd.; GB 402826	Standard Brands Inc.; US 1984789; filed December 7, 1932		Antifoam distribution
December 9, 1932, July 11, 1933, and September 19, 1934 (AT)	Vogelbusch; AT 142217; DE 682136	Vereinigte Mautner Markhof'sche Presshefe Fabriken; AG Ignaz Kuffner & Jacob Kuffner für Brauerei, Spiritus- und Presshefefabrikation Ottakring-Döbling; AG für Spiritus- und Presshefe-Industrie Wolfrum; M. Fischl's Söhne; Oberösterreichische Spiritus- & Presshefefabrik Josef Kirchmeir & Sohn, Brueder Reininghaus AG für Brauerei- und Spiritus-Industrie; AT 148970; filed July 27, 1933		Air distribution
August 23, 1935 (GB)	Lockey; The Distillers Co. Ltd.; GB 461439	Standard Brands Inc.; US 2151126; filed July 24, 1936		Air distribution
November 22, 1935 (DE)	Bergius and others (1939); Deutsche Bergin AG Für Holzhydrolyse; DE 659333	N. V. Intl. Suiker en Alcohol Compagnie Intl. Sugar and Alcohol "Isaco"; US 2146326; FR 813790; GB 469300; first filed November 17, 1936		Air distribution
October 12, 1945 (US)	Liebman and de Becze; Schenley Industries, Inc.; US 2530814	Farbenfabriken Bayer AG; DE 920844; filed July 13, 1950	Late filing date, probably extended due to the war	Air distribution
October 19, 2000 (NL)	Broecke and others; DSM N.V.; WO 0233048	Lesaffre et Compagnie; AT 473270; DK 1341898; ES 2350971; PT 1341898; filed September 26, 2001		Air circulation

^aWhen identical versions of the same "basic patent" were issued in several countries, patent numbers are given in parentheses and without reference to the publication year. Identification of the country codes is given in Table 2.

Table 5—Patents published between 1900 and 2014 on the control of aeration during the production of baker's yeast.^a

Inventor (publication y); priority date	Country	Patent nr.	Details
Braasch (1911); March 31, 1909 (DE)	DE	GB 191007640 (not accepted)	Under pressure, the full amount of air is introduced until the end of the fermentation; yield
Radiotechnische Studiengesellschaft mbH (1914); May 30, 1912	DE	DE 275432 (AT 67438)	A tube containing little radioactive material is immersed into the medium, and forms oxygen, avoiding aeration
Meyer (1920); May 31, 1919	NL	NL 8368 (FR 505907; GB 144244)	Without direct aeration and under reduced pressure, liquid is agitated to continuously recover alcohol and carbon dioxide; air is said to be nonessential for yeast growth
Jansen and Jansen (1928); May 3, 1927	NL	GB 291127	Nutrients (phosphates, ammonium, acids, alkalis, and so on) in a very finely divided spray are supplied through the aeration system; yield
Norddeutsche Hefeindustrie AG (1934); July 14, 1928 ^b	DE	DE 590550	Aeration rate is adjusted according to yeast cell growth; yield and energy costs
Stich (1932); September 25, 1928 ^b	DE	DE 553231	Individual elements of the aeration system are provided with separate control
Braasch and Braasch (1938); licensed to Standard Brands Inc. (US and CA only); January 16, 1932 ^b	DE	DE 655729 (FR 750158; GB 404459; US 2013456; CA 361390)	Carbon dioxide (1% to 3%) or waste fermentation gas is introduced into the air system; to prevent infection and lowering of the yeast activity, and to improve its yield and color
Gyri and Mayrhofer (1935); October 9, 1933	AT	AT 140462	Waste fermentation gas is returned to the vat, after recovering carbon dioxide
Kokatnür (1940); assigned to Autoxygen Inc.; December 9, 1936	US	US 2218336	Vat, aeration system or cloth bag is coated with a layer of fatty acid peroxides, peracids and salts from cocoa butter or stearic acid; peroxides liberate oxygen; yield, protein content, odor, color, stress tolerance, and viability of baker's yeast
Rosenqvist (1940); assigned to Svenska Jästfabriks Aktiebolaget; January 22, 1938 (SE) ^b	SE	DE 691862 (US 2281457; FR 857337)	Air delivery is controlled by a determined program; yield
Dorda (1940); assigned to Hefe-Patent GmbH; March 14, 1939	DE	DE 700492	Air is supplemented with ozone according to a predetermined program based on yeast cells concentration; yield, color, and keeping properties
Olsen (1954); February 19, 1952	GB; DK	GB 710532	Air valve capable of withstanding moderate pressures, and adapted to be controlled remotely
Kautzmann (1961); assigned to Hefe-Patent GmbH (1961); October 10, 1959 ^b	DE	DE 1107172 (AT 224053; FR 1269543; GB 923286)	Continuous measurement of dissolved oxygen with electrodes connected with a regulating device for metering air feed
Schulz (1966); November 16, 1962	DE	DE 1226064	Continuous process based on foaming and control with specific floaters and plate drums
Svenska Jästfabriks Aktiebolaget (1964; 1965); February 5, 1963 (SE) ^b	SE	FR 1379225 (GB 1035552); FR 1390262 (GB 1035551)	Using an apparatus for analyzing alcohol and dissolved oxygen (electrode), feedback control of the oxygen transfer is performed by varying the rate of rotation of the agitator and/or the surface tension of the medium with antifoam
Ebner and Enenkel (1971); assigned to Heinrich Frings; April 8, 1970 (AT)	DE	DE 2109896 (JPS 5546707; ZA 7102152)	Using a specific aeration device (AT 229083), excess aeration is provided for only 3 to 4 h, then aeration is kept constant, and molasses feeding is regulated by volatile components in exhaust (0.01%); close to fermentation control
Ishikawa and others (1976); September 3, 1973	JP	JP 51009833	Mixture of air and pure oxygen; yield
Midorikawa and Sasanami (1974); Dainippon Ink and Chemicals, and Japan Gasoline Co. Ltd. (1975); January 9, 1974 (US)	JP	US 3846246 (FR 2258447)	Growth medium is aerated at 14 to 32 cm/s instead of 7–10 cm/s (standard aeration rate for baker's yeast) to avoid foaming
Müller (1975); February 12, 1974	CH	CH 586350 (GB 1486488; DE 2505747; DK 49475; FR 2308782; SE 7500384)	Apparatus for reducing the power required for aeration; residual energy from exhaust gas is recovered
Buchholz and others (1989); assigned to Hoechst AG; November 23, 1987	DE	DE 03739621 (EP 0317854; CA 1312568)	Oxygen is transported with the fluid carrier; carbon dioxide is displaced, desorbed, and removed to avoid gas waste

^a Linked to the inventor and publication year (left column), patent number refers to the earliest publication of the invention. Patents are listed according to filing date of the invention (priority). For old German patents, the priority date was based on the information disclosed in the publication ("Patentiert vom"), which was 1 d later than the "convention date" (filing date) cited in foreign patents; when the original German patent was not found, the convention date reported in foreign patents was used. When available, country of filing is mentioned only when different than the main patent. For some entries, details of the main invention were published in several "basic patents" forming a "patent family." When identical versions of the same "basic patent" were issued in several countries, patent numbers are given in parentheses and without reference to the publication year. Within each entry, the filing date determines the order of presentation of patent numbers. Identification of most of the country codes is given at the bottom of Table 2.

^b Breakthrough.

Table 6—Patents published between 1900 and 2014 on air circulation (inlet) during the production of baker's yeast.

Inventor (publication y); priority date	Country	Patent nr.	Details
Giesecke (1914a, 1914b, 1914c); June 3, 1911, June 24, 1911, and August 6, 1912	AT-HU (AT)	AT 65676; AT 65677; AT 65678	Includes a suction apparatus at the bottom
Bratton (1929); assigned to Anheuser-Busch, Inc.; March 6, 1926	US	US 1732921	Air escaping is passed several times through the fermentation medium
I. G. Farbenindustrie AG (1929); November 7, 1927	DE	GB 309207	Within a closed vessel containing a series of trays, liquid runs slowly on a shallow surface and in opposite direction to air flow
Jansen (1932, 1935); licensed (DE only) to Norddeutsche Hefeindustrie AG (1935, 1938; no cross references); February 20, 1930, and December 6, 1933 (NO)	NO	DE 621566 (AT 138613; DK 47315; DK 47373; GB 366753; FI 16624; DE 666321); US 2056576 (FR 782281)	Medium is subjected to violent agitation, withdrawn with exhaust gas (to allow continuous recovery of alcohol and carbon dioxide), and returned to the main vat; yield
Vogelbusch (1931); March 22, 1930	AT	AT 123393	Liquid circulates and is aerated outside of the main vat
Simmer (1931); April 19, 1930	PL	AT 123813 (DE 577244)	Vat is divided into 6 to 20 compartments to improve control over aeration and other parameters
Nilsson (1933); Svenska Jästfabriks Aktiebolaget (1936); December 23, 1930	SE	SE 77912 (DK 47925; DE 637728)	Liquid circulates into 2 U-shaped columns where part of it is aspirated, and returned with compressed air into the mainstream
Jonáš (1938); January 18, 1934	CS	DE 657547 (AT 145692)	Portion of the liquid is withdrawn, emulsified with air through a centrifuge, and returned in the vat through a perforated tube network
Effront and Popper (1935); Effront (1936); April 7, 1934	FR; HU	FR 784127 (GB 451062; DE 656361; US 2123463)	Air circulates into a series of vats equipped with an ascension conduit
Bergius and others (1939); assigned to Deutsche Bergin AG für Holzhydrolyse, and others (1938); licensed (US, FR, and GB) to N. V. Intl. Suiker en Alcohol Compagnie Intl. Sugar and Alcohol "Isaco"; November 22, 1935	DE	DE 659333 (US 2146326; FR 813790; GB 469300)	Air is supplied to the fermentation liquid which trickles down a tower over filling bodies
Stich (1939); Stich and Kottlors (1939); December 23, 1937 (DE)	DE	US 2244902 (DK 59951; GB 509745; FR 849162)	Liquid circulates through a number of containers arranged one after another from top to bottom and vice versa, and supplied with repeated contact with air; for large consumers of fresh yeast desiring to produce it themselves
Stich (1944, 1953); September 16, 1938, and December 28, 1941	DE	DE 743984; DE 754029	Air is pushed from the top down through the liquid
Claus (1944, 1952, 1954); assigned to Zellstoffabrik Waldhof; April 5, 1940, March 11, 1941, and October 19, 1941	DE	DE 759121 (DK62691; FR 871295); DE 905965; DE 744678 (DK 65916; BE 44347)	Emulsified liquid is aspirated from the top, and recirculated through the aeration unit where it is directed into a closed chamber and defoamed
Vogelbusch (1943, 1953); November 12, 1940 (DE), January 9, 1941 (DE), and August 23, 1945 (AT)	AT	DE 738961 (FR 878327); DE 869786	Foam is de-aerated with a defoamer, and returned to the bottom of the fermentation vat through a specific aeration device providing fine air bubbles
Stich (1953a, 1953b, 1943, 1944, 1953c, 1953d); January 9, 1941, January 9, 1942, March 19, 1942, June 6, 1942, June 24, 1942, and February 10, 1943	DE	DE 752635; FR 893089 (DK 67453); DE 752721 (FR 894327; DK 66421); DE 744176 (FR 893052; DK 68006); DE 752360 (FR 894226; DK 66662); DE 752312 (FR 900839)	Liquid is aerated, and de-aerated outside of the main vat; as part of a continuous process in closed circuit
Källner and others (1954); assigned to Phrix-Werke AG; May 9, 1942	DE	DE 911962	Supplementary air system is situated above the surface of the liquid, provides air downward, and sucks foam from the middle section of the device; to prevent foaming
Stich (1954); August 30, 1942	DE	DE 751317	The 1st air system is at the bottom of the tank; the 2nd system is at some distance from the bottom and below a funnel which allows the liquid to be brought over the surface of the foam (to destroy it)
Humfeld and others (1951); assigned to the Secretary of Agriculture, the United States; September 12, 1947	US	US 2542031	Vessel includes lateral passageways communicating with the main chamber, upwardly and outwardly; with a foam breaker (patented separately, US 2610155)
Stich (1962); July 28, 1960 (DE)	DE	CH 369418 (FR 1287771; US 3114677)	Aeration device is placed outside and under the fermentation vat; especially for continuous processes

(Continued)

Table 6—Continued.

Inventor (publication y); priority date	Country	Patent nr.	Details
Frings GmbH (1968, 1975); Ebner and Enenkel (1971, 1974); April 5, 1966 (AT), and September 1, 1972 (DE)	DE	AT 269083 (GB 1135878; US 3813086; CH 454804; SE 320041; DE 1667042; DE 2243161; DE 2244284); AT 319865 (US 3891729 and 13 others)	Self-aspirating aerating device is disposed into a vertically disposed tank; reduced air consumption and improved yield
Brooks and Nubel (1969); assigned to Chas. Pfizer & Co., Inc.; December 29, 1966	US	US 3476366 (FR 1553995; BE 708543; GB 1199779; DE 1667231)	Aeration device is located close to the upper surface of the liquid which is withdrawn from the bottom, and returned to the top of the hollow vertical elongated vessel
Vogelbusch GmbH (1971); Vogelbusch (1974); Müllner and Vogelbusch (1973; SE only); November 20, 1968, and July 11, 1972	AT	AT 288294 (FR 2023686; GB 1294466; DE 1957160; CH 522431; NL 6917453; SE 353739); AT 317122	Air is introduced from the top, and meets specific deflectors
Vogelbusch GmbH (1977a, 1977b); July 14, 1975	AT	AT 339847; AT 339848	Liquid is injected into a separate aeration unit installed at the bottom of the vat where it is strongly mixed; yield
Bernstein and others (1989); assigned to Fried. Krupp GmbH; April 20, 1988	DE	EP 0339356	Under partial pressure, the liquid is held in forced circulation in a loop reactor, the velocity of the medium being much greater than the rate of ascent of the gas bubbles
Speece (2001); assigned to Eco-Oxygen Technologies, LLC; October 2, 2000	US	US 6322055	Gas is dissolved into a fluid which circulates in a specific manner at the bottom of 2 vertically oriented tubes
Broecke and others (2002); assigned to DSM N.V.; licensed to Lesaffre et Compagnie (AT, DK, ES, and PT only); October 19, 2000	NL	WO 0233048 (and 14 others)	Dual aeration system; injection of a first oxygen-containing gas (such as air) in a heterogeneously flow near the bottom of the fermenter, and introduction of a 2nd oxygen-containing gas (such as pure oxygen) as a heterogeneous flow of gas bubbles of nonuniform size

(1957), Jagusch and others (1969), and Jagusch and Schönherr (1974).

Air distribution

Compared to air control or circulation units, inventors had more interest in air distribution or diffusion into fermentation vats probably because it had the most critical effect on the growth of yeast biomass (Table 7). In early designs, a screen was placed over the aeration unit at the bottom of the fermentation tank and this was later replaced with perforated tubes. From 1920 to the 1950s, pipes were gradually equipped with the finest and most efficient distribution units available such as porous stones or a diaphragm. However, there was a limit to the size of the holes because it would clog, so cleaning of such aeration units was also a concern for inventors. In most cases, the proposed inventions on air distribution in liquid fermentation media relied much on excellent mixing devices to break air bubbles and distribute them in the most efficient manner so air would remain as long as possible in the liquid column. Conditions that result in improved mixing of liquid media are likely to improve yeast yield, although increased agitation as such would have a minor effect (West and Gaden 1959). Many styles of air diffusers were invented. In recent years, new concepts of aeration units have been proposed for miscellaneous industrial applications. This suggests that existing aeration units are still not considered optimal for baker's yeast production.

Historical context

Practiced since the end of the 1870s, the decision to aerate yeast to stimulate its growth was a major breakthrough because it increased much the yield of baker's yeast. As mentioned, inventors showed great interest for aeration when both molasses-based media and fermentation controls were improved (Figure 2). In Germany,

strong aeration appears to have been a popular practice around 1895 to 1900. On the contrary, Distillers Co. Ltd. (DCL), the largest yeast manufacturer in the United Kingdom, did not adopt the aeration process until 1919 (Weir 1991). According to this author, Fleischmann Co. from the United States provided technical skill on the aerated process through the formation of Intl. Yeast Co., a new company based in Great Britain. This example illustrates the major differences in the early development of industrial yeast manufacturing, depending on the country.

Despite high energy costs, aeration reduced the cost of commercial baker's yeast because much more biomass could be produced from the same quantity of molasses. Soon, there was a need to optimize the design of aeration units and reduce energy and avoid foaming. According to some authors (de Becze and Liebmann 1944; Irvin 1954; Pyke 1957), aeration units, such as the porous-type developed by Stich and Vogelbusch, were mainly popular in Europe, but not in the United States due to high maintenance costs and cleaning problems.

Compared to anaerobic yeast cells harvested from alcohol manufacturing, keeping properties of aerated yeast cells were also improved, partly due to their lower enzymatic activity and higher glucose-based reserve materials. However, several inventors mentioned that the gassing power of so-called "air-yeast" was much reduced when incorporated into bread dough which is anaerobic by nature. In many patent specifications, yeast grown under conditions allowing some alcohol production were said to have superior gassing power compared to air-grown yeast. According to Effront and Popper (1936), "If yeast is fermented in alcoholic media, as is the case for the old "Viennese Process," high enzymatic strength develops under natural conditions, and yeast incorporated into dough keeps great strength if its nutrition has been correctly regulated. But yeast preparation according to this process is not

Table 7—Patents published between 1900 and 2014 on air distribution (diffusion) during the production of baker's yeast.

Inventor (publication y); priority date	Country	Patent nr.	Details
Barbet (1900); August 8, 1899	FR	GB 189916159 (US 668080)	A series of trays or shelves aerate thin layers of circulating media by forming tiny bubbles; sterile
Klipfel (1901); January 2, 1901 ^a	CH	CH 21295	Air is introduced at the bottom of the vat through a series of perforated and revolving tubes; for beer and baker's yeast
Strauch (1912a, 1912b, 1912c); December 13, 1910, March 3, 1911 (DE), and November 4, 1911	DE	DE 247488 (AT 53997); AT 53999; DE 250735	Perforated tubes are placed in individual boxes; to facilitate cleaning
Maschinenbau AG Golzen-Grimma (1912); June 21, 1911	DE	DE 246709	Mobile (back and forth) system with perpendicular branches placed at the bottom of a rectangular vat
Ullrich & Hinrichs AG (1913); August 5, 1911	DE	DE 257557 (AT 54280)	One or several screens installed over the aeration device
Hafen (1914); December 31, 1912	DE	DE 271250	Liquid is repeatedly mixed outside of the vat, and foam is returned through the perforated bottom of the vat
Pick (1922); May 18, 1916	CS	AT 88640	Liquid is finely sprayed, and mixed countercurrent with air
Peter (1921a, 1921b, 1921c); September 25, 1917, November 7, 1917, and February 2, 1918	DE	DE 336246; DE 338886; DE 345552	Air or oxygen is introduced at the bottom of the vat through porous tubes, under a mixing device that divides bubbles in very fine particles; yield, and defoaming
Fehrmann (1923); April 27, 1921	DE	DE 376464	Sections of the grid are detachable, and have specific grooves and uneven surfaces
Bryant (1922, 1923); May 18, 1921, and August 11, 1922	GB	GB 184860; GB 194623	Minute bubbles of air are introduced through perforated pipes containing porous stones or a gauze envelope filled with sand between perforated sheets
Strauch & Schmidt (1932, 1934); July 21, 1925, and October 10, 1925	DE	DE 552241; DE 597953	Distribution pipes perpendicular to the main pipe, with shutoff valves
Jansen (1927); November 10, 1925	NL	GB 270770	Aeration unit providing very small bubbles (0.1 to 0.2 L per min) to a thin film of growth medium
Stich (1933a, 1933b); licensed to Norddeutsche Hefeindustrie AG (1934a, 1934b, 1934c, 1934d, 1934e); January 28, 1927, October 25, 1927, June 11, 1929, December 28, 1929, August 5, 1929 (AT), June 20, 1930, and December 13, 1930 ^a	DE	DE 567518; DE 594361; DE 594192; DE 594193; AT 131579 (FI 14741; GR 2914); DE 594194; DE 594195	Air diffuser; froth of fine air bubbles is slowly formed through double-plate diaphragm plates such as those used in electrolytic cells, with bottom layer (coarse pores) and upper layer (fine pores; 0.16 to 6 μm); higher yield with less air; 1st patent on the design of aerating pores
Braasch and Braasch (1930); October 17, 1927 (DE)	DE	AT 117459	Air diffuser; holes are coated with rubber
Stich (1934); licensed (AT only) to AG Ignaz Kuffner & Jacob Kuffner für Brauerei, Spiritus- und Presshefefabrikation Ottakring-Döbling, AG für Spiritus- und Presshefe-Industrie Wolfrum, M. Fischl's Söhne und Vereinigte Mautner Markhof'sche Presshefe Fabriken (1933); October 25, 1927, and August 5, 1929 (AT)	DE	DE 594361; AT 134100 (FR 680303; DK 47748; US 1792450; GB 340644)	Several closed fermenting compartments with increasing volume to allow better aeration control, using diaphragm plates described by Stich in DE 594361 (above entry); continuous process
Stich (1931a, 1932, 1931b); January 5, 1929, December 28, 1929, and January 7, 1930	DE	DE 507294; DE 561978; DE 526951	Air diffuser; individual elements of the system are made of flexible materials (rubber) to prevent entry of liquids
Norddeutsche Hefeindustrie AG (1934); Wirtschaftliche Vereinigung der Deutschen Hefeindustrie (1935a, 1935b, 1936); Stich (other than DE); January 5, 1929, January 7, 1930, February 25, 1930, and May 29, 1931	DE	DE 594671; DE 622962; DE 622963 (FR 709948; CH 152237; AT 133117; US 1959554; DK 48887); DE 627549	Fine air bubbles are slowly provided through thick-walled hollow aerating bodies having pores about 25 μm or less, with inside diameter being larger than outside diameter; valves regulate inlet air; preferable to smaller pores (difficult to manufacture and resistant to air passage)
Technisches Büro Vogelbusch (1931, 1932); October 3, 1929, and March 8, 1930	AT	AT 122951; AT 127365	Air diffuser; corrugated sheet with holes of small diameter

(Continued)

Table 7—Continued

Inventor (publication y); priority date	Country	Patent nr.	Details
Bohm (1934); December 6, 1929	DE	DE 602743	Device allowing pivoting of the pipes on the walls of the vat to facilitate cleaning
Vogelbusch (1932); April 26, 1930	AT	AT 128825	Air diffuser; tubes are covered by a screen (metal or tissue such as hemp, jute, cotton, or silk)
Strauch & Schmidt (1935); May 14, 1930	DE	DE 621356	Beam-lines lying on several levels are connected to chambers
Vogelbusch (1932); July 25, 1930	AT	AT 126573	Air diffuser having holes of specific shapes; easier to manufacture
Boidin and Effront (1931); July 28, 1930	FR	FR 714644 (DE 561724)	Several horizontal plates rotate
Strauch & Schmidt (1931); October 10, 1930 (DE)	DE	FR 710667	Air diffuser with small holes (0.2 to 0.4 mm)
Bermann (1935); assigned to Verband der csl. Presshefefabriken; December 10, 1930	CS	CS 51832	Air diffuser; glass filter plates with a pore size of 40 to 70 μm
Stich (1932a, 1932b); January 16, and April 30, 1931	DE	DE 543776; DE 545345 (DK 47982; AT 132390; CH 160162; FR 729596; GB 376645; US 1920719; CA 331971; FI 15449)	Air is diffused through a hollow body with small perforations which close when air pressure drops, to prevent entry of liquids into the aeration system
Bohm (1932); January 30, 1931	DE	DE 564496	Air diffuser; elastic intermediate lines connect the air supply lines to the diffusion bodies
Fuchs (1932); March 6, 1931	DE	DE 552978	Mobile air diffuser incorporated into a cylinder and easy to clean
Sinner AG (1936); November 26, 1931	DE	DE 629884	Air diffuser includes a pneumatic manifold which changes air bubble size
The Distillers Co. Ltd. and Locket (1933a, 1933b, 1935); Locket (1938, 1939); licensed to Standard Brands Inc. (US only); January 7, 1932, February 21, 1933, and March 27, 1935	GB	GB 387486 (US 2104330; US 2176558; GB 393551; GB 431674)	Air diffuser comprises a hollow body with small perforations enveloped by a coil; to prevent clogging, and to improve cleaning
Braasch and Braasch (1933); Braasch (1934); licensed to Standard Brands Inc. (US only); January 10, 1932	DE	DE 605912 (GB 395610; FR 748594; US 2092151)	Small quantities of organic acids (lactic or acetic) are introduced into the air system by means of a diaphragm having a very fine porosity (less than 6 μm); to reduce the surface tension of the fermenting liquid, and give finer bubbles
Madsen (1934); January 21, 1932	DK	DK 48730	Air is provided through a revolving helix (perforated)
Industriemiska Aktiebolaget (1933); Wessblad and others (1937, 1938); January 30, 1932 (SE)	SE	FR 749848 (GB 414117; AT 140195; US 2077652; US 2118370)	A thin layer of fermenting liquid is circulated over a perforated screen
Maschinenbau AG Golzern-Grimma (1934); September 9, 1932 (DK)	DE	FR 764873 (DE 629194)	Through horizontal bars, air is introduced at different levels of the tank and under adjusted air pressure conditions
Norddeutsche Hefeindustrie AG (1935); September 26, 1932 (DE)	DE	GB 423331	Propeller or screw equipped with an air conduct and arranged to impart a strong horizontal helical streaming motion
Vogelbusch (1934); September 30, 1932	AT	AT 136969	Air diffuser; small juxtaposed rings
Gosda (1934); November 22, 1932 (DE)	PL (Danzig)	GB 412842	Porous pipes branch from a central air supply pipe and extending in grid shape over the bottom of the vat
Vogelbusch (1935, 1939, 1936); December 9, 1932, July 11, 1933, and September 19, 1934 ^a Licensed to (AT 148970 only): Vereinigte Mautner Markhof'sche Presshefe Fabriken; AG Ignaz Kuffner & Jacob Kuffner für Brauerei, Spiritus- und Presshefefabrikation Ottakring-Döbling; AG für Spiritus- und Presshefe-Industrie Wolfrum; M. Fischl's Söhne; Oberösterreichische Spiritus- & Presshefefabrik Josef Kirchmeir & Sohn, Brüder Reininghaus AG für Brauerei- und Spiritus-Industrie	AT	AT 142217; DE 682136 (AT 148970; US 2121458; DK 50578; FR 765295; GB 430348; GB 431566); AT 148020	Air diffuser with large openings; air is allowed to escape at right angle through holes with an aerodynamic shape, then bubbles are broken; a nutritive solution may be mixed with air; to reduce the amount of air

Table 7—Continued

Inventor (publication y); priority date	Country	Patent nr.	Details
N. V. Industriele Maatschappij v. h. Noury & van der Lande (1934); van Thiel (1936); January 11, 1933 (NL)	NL; FR	FR 766505 (GB 431688; US 2043940)	Large bubbles are divided very finely with a screen
Scholler and Eickemeyer (1936); Scholler (1936); January 11, 1933	DE	DE 744133 (GB 448689; US 2083347; US 2083348; FR 799358)	Aeration device (conical); liquid is streamed in vertical directions, within a system of vertical and inclined surfaces
Rosenqvist (1937); assigned to Svenska Jästfabrik Aktiebolaget; October 10, 1933	SE	SE 88558	Liquid is agitated over and around the air inlet
de Becze (1934); unknown filing date	HU	HU 110202	Air passes through perforated tubes or plates such as a reversed funnel
Schattaneck (1935); March 14, 1934	AT	AT 147020 (FR 787817; DE 648630; US 2038451)	Air diffuser; a metal plate made of copper alloy which is easy to manufacture, and to limit infection
Drake and Gardella (1937); August 29, 1934, and December 5, 1935	US	US 2061564 (GB 475076; CA 364845)	Impeller blades are inclined to form specific cavitation pockets in the fluid
N. V. Industriele Maatschappij v. h. Noury & van der Lande (1936); van der Lande and van Thiel (1938); December 29, 1934 (NL)	NL; FR	FR 800109 (GB 472622, not accepted; CH 186840; US 2119188)	Air-in-liquid dispersion is prepared by centrifuging a dilute molasses solution with a small amount of toluol; to improve oxygen transfer and reduce air quantities
The Distillers Co. Ltd. and Lockey (1937); Lockey (1939); licensed to Standard Brands Inc. (US only); August 23, 1935	GB	GB 461439 (FR 809706; US 2151126)	Gas diffuser (central) consisting of a perforated conical device which rotates, and creates vigorous turbulence
Drake (1937); October 28, 1935	US	US 2094004	Liquid is sucked from the bottom, and is circulated upward by an impeller (upper portion of the housing is cone-shaped); improved air diffusion
Kissling (1937); November 14, 1935	AT	AT 149901	Vat is sealed, and pressurized to increase the concentration of dissolved oxygen; yield
Rank (1937); May 11, 1936	DE	FR 808293 (US 2255532; DE 734072)	Air diffuser; holes made of wires having small acute angle gas passages; easy to clean
Drake (1939); February 1, 1937	US	US 2148445 (GB 508385; CA 380436)	Device includes a specific impeller that maintains the liquid in continuous circulation
Sinner AG (1939); August 23, 1937, and April 21, 1938 (DE)	DE	FR 837743 (GB 503165; AT 160214)	Air diffuser; rotating, bell-shaped, and without baffle plates
Opderbeck and Koch (1955); assigned to Phrix-Werke AG; October 3, 1944	DE	DE 928224	Air is introduced below a bell-shaped device which allows the passage of air bubbles at the top
Hatch and Hammond (1949, 1950); assigned to Weyerhaeuser Timber Co.; May 16, 1945, and March 27, 1947	US	US 2482908; US 2522947	Air is beaten and agitated into a central portion of the vat, forming much foam which is re-circulated; yield
Liebman and de Becze (1950); assigned to Schenley Industries, Inc. (1953); licensed to Farbenfabriken Bayer AG (DE); October 12, 1945	US	US 2530814 (DE 920844; FR 1026775)	Aeration device; air is distributed with a specific impeller with baffles; for several microorganisms
Nordell (1951, 1953); February 26, 1947, and January 27, 1951	US	US 2555210; US 2650810	Air diffuser comprises a hollow core having a plurality of orifices and a covering of compressible monofilament cord coiled tightly around said core; support is also provided for raising and lowering diffuser tubes
Hilbers and Junge (1953); October 2, 1948	DE	DE 862739	Air distributor avoids the formation of large bubbles at the top of the vat
Streckert (1953); assigned to Sinner AG; January 11, 1951	DE	DE 900087	Rotating blower consisting of a hollow body made of 2 chambers in which air bubbles are broken

(Continued)

Table 7—Continued

Inventor (publication y); priority date	Country	Patent nr.	Details
Philosophow (1953); partly assigned to Zellstoffabrik Waldhof; February 23, 1951	DE	DE 879081	A rotating half-cone at the bottom of the vat allows both air addition and aspiration
Tapleshay and Nordell (1954); Chicago Pump Co. (1955); August 25, 1951	US	US 2694041 (GB 721715)	Air diffuser, with prehumidified air to prevent clogging
Vogelbusch (1953, 1954); April 7, 1952, and October 11, 1952	AT	AT 175861; AT 179281	Air diffuser; air is distributed from a specific perforated hollow shaft over a rotating blade
Olsen (1954a, 1954b); December 18, 1951	GB; DK	GB 718292; GB 718293	Nozzle is mounted on a telescopic tube; gas chamber is arranged with removable covers at top and bottom; easy-to-clean
Vogelbusch (1953); April 15, 1952	AT	AT 175862	Air bubbles are propelled at the bottom of the vat
Stimpson and Trebler (1956); assigned to Natl. Dairy Research Laboratories, Inc.; March 20, 1952	US	US 2750328	Includes a rotatable impeller and a vertical open-ended tube
Schaller (1956); assigned to Phrix-Werke AG; November 15, 1953	DE	DE 953691	Cone-shaped and spiral mixer; less foam
Vogelbusch (1956); January 27, 1955	AT	AT 186596 (DE 1716318)	Vat is cleaned with a rotating device which is part of the aeration system; improved recovery of yeast and cleaning
Stich (1959); May 2, 1955	DE	DE 1064458	Air diffusers; a series of porous tubes and casings giving very fine air bubbles
Patentauswertung Vogelbusch GmbH (1957); July 8, 1955	AT	AT 188676 (FR 1156344)	Both small and large bubbles are formed simultaneously, and introduced independently at different points of the tank
Patentauswertung Vogelbusch GmbH (1958a, 1958b, 1960a, 1963, 1960b, 1962, 1963, 1964a, 1964b, 1962); October 8, 1956, February 14, 1957, April 20, 1957, April 20, 1957, April 25, 1957, December 19, 1957, February 18, 1959, May 16, 1959, November 23, 1960, and December 1, 1960	AT	AT 198734; AT 199170; AT 207369 (DE 1189522); AT 225162 (DE 1191339); AT 212288 (DE 111320); AT 219567; AT 230857; AT 236924 (DE 1190435); AT 234651; AT 221075 (DE 1193478)	Air diffusers (various types); air is distributed with an impeller having an aerodynamic shape
Thommel and Brumme (1959, 1960); assigned to Hefefabrik Weingarten GmbH (DE only) and to Escher Wyss GmbH; May 27, 1957, and August 8, 1958	DE	DE 1071024; DE 1083771 (GB 912176; CH 379445; US 3066921)	Vertical hollow air supply shaft is mounted for rotation in a container, and a plurality of air outlet openings is arranged to avoid liquid backflow
Patentauswertung Vogelbusch GmbH (1959a, 1960a, 1960b, 1959b); July 27, August 14, August 17, and October 23, 1957	AT	AT 204015 (DE 1124018); AT 209879; AT 208343 (DE 1137420); AT 202117	Rotating unit for the fine distribution of gas; several variations exist
Stich (1962); assigned to Gebr. Becker Apparatebau (1960); March 1, 1958 (DE)	DE	US 3068155 (FR 1221178; CH 380684; AT 226180; GB 899478)	Diaphragm rendering possible working with high yeast concentrations (up to 50%), especially suitable for the continuous process
Patentauswertung Vogelbusch GmbH (1960); Braun (1963); April 29, 1958, and June 19, 1959	AT	AT 212287 (DE 1196165; GB 877652; US 3092678; CH 369749; FR 1222291; DK 92538); AT 219566	Gas distributing device rotates, and forms large and small bubbles; liquid is circulated rapidly to the point where it is replenished with gas
Goos and Florin (1967); assigned to Escher Wyss GmbH; May 30, 1962	DE	DE 1244121	Device allowing the creation of a vortex
Weinrich (1966); August 7, 1962	DE	DE 1230399 (FI 41537)	Air diffuser consisting of a plurality of rotating elements
Kersting (1966); October 15, 1963	DE	DE 1232913 (GB 1032189; US 3274075)	As part of a continuous process, aerating elements radiate spirally from a central air-feed pipe, and are set at a particular angle of incidence to the longitudinal axis of the tank (obliquely ascending); yield
Patentauswertung Vogelbusch GmbH (1966); Hofschneider (1968); May 8, 1964	AT	AT 247823 (DE 1230398; FR 1450517; US 3400051; GB 1035868; CH 431463; SE 304739)	A rotating gas injector is located below the vessel, and at least 3 vertically mounted baffles are connected to the vessel walls for preventing the fluid from rotating with the injector

(Continued)

Table 7–Continued

Inventor (publication y); priority date	Country	Patent nr.	Details
Höhne (1970); February 7, 1969 (DE)	AU	DE 1905998 (US 3829068)	A pump rolls the fluid over; stationary gas distributor includes pipes provided with gas outlets, with some variations depending on patent; alternative to rotating gas distributors; to reduce power consumption
Vogelbusch GmbH (1973); March 3, 1972	AT	AT 315139 (DE 2310319; CH 550020)	Aeration device equipped with a specific impeller; for culturing microorganisms
Vogelbusch GmbH (1975); Braun and Hofschneider (1974); November 15, 1972	AT	AT 326488 (FR 2206121; DE 2355639; GB 1416107; CH 563531)	Liquid is sucked and recycled through an aeration station equipped with a specific impeller; for fermentation media (yeast not specifically mentioned)
Vogelbusch GmbH (1977); April 30, 1973	AT	AT 335954	Liquid is sucked and recycled through an aeration station equipped with a specific impeller; for fermentation media (yeast not specifically mentioned)
Vogelbusch GmbH (1976); Braun and Hofschneider (1976); July 8, 1974	AT	AT 334210 (DE 2521931)	Liquid is sucked and recycled through an aeration station equipped with a specific impeller and a specific air diffuser; has steps in diffusor walls to break down large gas bubbles; for fermentation media (yeast not specifically mentioned)
Müllner (1977); Waagner-Biro AG (1978); July 7, 1975	AT	AT 336517 (GB 1515221)	Air is sucked by means of small diameter tubes which standardize bubbles size
Sonoyama and others (1980); assigned to Shionogi & Co. Ltd.; January 18, 1979	JP	JP 55097237 (GB 2041773; US 4287137)	Vane- or blade-type fluid impeller for various industrial applications including baker's yeast; lower cost
Bernstein and Zeitsch (1990); assigned to Fried. Krupp GmbH; August 9, 1988	DE	DE 3826955	Air is introduced in fine distribution from top to bottom
Höhne (1990); September 29, 1988	AU	DE 3833003	Outlets of air distribution pipes are underneath vertical running channel-like guidance chambers
Cahoon (1997); April 21, 1995	US	US 5616493	Liquid is maintained into a continuous foam, and yeast cells are recovered after breaking the foam; to eliminate mixing or stirring; economical
Höhne (1997); assigned to Burns Philp Food Holdings Pty. Ltd.; November 27, 1995 (AU)	AU	WO 9719747	Air diffuser; liquid nozzle having a functionally elongate outlet bounded on each side by a gas nozzle
Höhne (2000); November 4, 1999 (AU)	AU	WO 0027514	A liquid chamber creating much turbulence and minute gas bubbles
Luzhkov and Dzafarov (2008); September 18, 2006	RU	RU 2319381 (WO 2001035999)	Air bubbleers are at the bottom; additional bubbler is in the lower part of space between casing and circulation sleeve
Longman and others (2012); assigned to Acal Energy Ltd.; September 23, 2010	GB	GB 2484070	Device for generating fine bubbles (50 to 100 μm)

^a Breakthrough.

lucrative because, compared to a high alcohol production that is difficult to sell, yeast yield is low. For this reason, in modern processes, to avoid alcohol production, optimal dilution is necessary to feed continuously or periodically nutrients in combination with high aeration. High yeast yields are then attained, but yeast quality is not similar to “alcohol yeast.” Keeping properties are better, but yeast strength is lower because less enzymes could be formed.”

Antifoam

Table 8 presents patented inventions on foam control with mechanical devices. The use of chemical antifoams has been a more lasting solution to control foam in yeast fermentation vats. Specific chemical antifoam formulations are reported in Table 9. With only 16 patents, inventions on the distribution of liquid antifoams are presented in Table 10. In addition to those reported in Table 8, 9, and 10, more patents on foam control have been issued, especially after 1950. Out of the scope of this review, these patents were developed for a wide range of microbiological, medical, and chemical industries. They are not reported here because baker's yeast was not specifically mentioned. For example, chemical defoamers

are regularly proposed for fermentation processes in general, for example, by Matani and Tanaka (2014). Nonspecific froth sensors in fermenters have been described by McKinney (1959) and Redikultsev and others (1981).

Historical context

In the 19th century, yeast manufacturers were accustomed to harvest yeast floating on the surface of growth media. Around 1900, specific mechanical devices were needed to better control foam when aeration became a standard practice. Yeast molasses-based fermentation media have a relatively high surface tension and the liquid easily tends to become frothy upon intense aeration and mixing. According to many patent specifications, mechanical antifoams, or powerful jets of air or liquid over the foam, had low efficiency because aeration tended to split up the bubbles of foam into smaller bubbles without bringing about a separation of gas and liquid. Power consumption of early foam disintegrators was also considerable (Nauc  r 1937). Efficient mechanical antifoams systems have generally been used in combination with chemical antifoams (Hall and others 1973).

Table 8—Patents published between 1900 and 2014 on mechanical foam control during the production of baker's yeast.

Inventor (publication y); priority date	Country	Patent nr.	Details on foam treatment
Raben and Wrede (1901); March 23, 1900 ^a	DE	DE 11 6455	Removed with a device revolving over the surface of the growth medium; 1st patent on foam control
Dampf-Kornbrennerei und Preszhefe-Fabriken AG (1909); February 25, 1909	DE	DE 21 6979	Removed from the main vat, and treated separately with fat
Strauch (1913); August 24, 1910	DE	DE 259484 (AT 53998)	Entrained into a siphon, and liquefied in contact with fat-covered surfaces
F. Wulf AG (1914); October 3, 1912	DE	DE 276604 (AT 70639)	Air jets are injected directly on foam
Reik (1915, 1919); April 27, 1914, and April 25, 1916	AT-HU (AT)	AT 70615 (DE 300985); DE 313108	Removed by suction through a funnel placed under the foam, sent at the bottom of the fermentation vat, and recirculated through the aeration system
Peter (1917a, 1917b); October 24, 1915, and February 10, 1916	DE	DE 299435 (AT 79936; DK 21497); DE 300828	Aspirated, and returned into the liquid in a liquefied state
Giesecke (1918); May 9, 1916	AT-HU (AT)	AT 74450	Removed with screen surfaces with the lower edges immersed
Verein der Spiritus-Fabrikanten in Deutschland (1919); Hayduck (1923); licensed to The Fleischmann Co. (US and CA); August 20, 1916	DE	DE 303801 (GB 155290; US 1449114; CA 233379)	Propelled at a high velocity, and liquefied; cheaper than fats
Vereinigte Nord- U. Süddeutsche Spiritwerke und Preszhefe-Fabrik Bast AG (1919); October 27, 1917	DE	DE 31 5957	Rust-like panels act as displacing baffles, the upper to the lower regulating the size of the outlet
Fischl (1920); May 1, 1918	AT	AT 80068	Beaten over the surface
Strauch & Schmidt (1919, 1921); August 17, 1918, and March 21, 1919	DE	DE 316441; DE 341108	Liquefied by spreading over a large surface (with air injections), and returned to the vat
Strauch & Schmidt (1922); November 13, 1919	DE	DE 362375	Sucked into towers over the growth medium, and liquified
Noe (1922); December 17, 1919	US	US 1402248	Deflected with a conical device, moved to a separate tank, and returned to the bottom of the main tank
Strauch & Schmidt (1922); May 25, 1921	DE	DE 364964	Seized by impact arms, compressed, broken, sucked, propelled on a surface, degassed, and pumped back into the tank
Hartmann (1922, 1924); July 16, 1921, and July 9, 1922	DE	DE 368574 (FR 544725; NL 11944); DE 402085	Sucked, and destroyed with an impeller
Vogelbusch (1924); January 4, 1922	AT	DE 395501	From under the foam, liquid is pumped, and finely spread over the foam
Sinapius (1924); February 25, 1922	DE	DE 388178	Beaten with a specific impeller placed on top of the vat
Norddeutsche Hefeindustrie AG (1929); December 19, 1926	DE	DE 483329	Growth medium is passed through active charcoal which absorbs foam; preferable to antifoam chemicals which affect the quality of the yeast, and are difficult to remove from tank surface
Stich (1935); February 12, 1933	DE	DE 621647	Sucked, and transferred counterflow to the liquid media and to a series of connected vessels
Nauclér (1937); June 14, 1935 (SE)	SE	GB 469058 (US 2184195; DE 733296)	Quenched into a separate container through a pressurized spraying device
Hefe-Patent GmbH (1938); May 31, 1936	DE	DE 657718	Additional aeration with large bubbles
Seidel (1953); September 4, 1937, and October 3, 1938	DE	DE 746731 (FR 878985)	Moved to a series of recipients, centrifuged, and returned to the vat; FR patent also covers nitrogen control
Vogelbusch (1942a, 1942b, 1942c, 1942d, 1943a, 1943b, 1943c, 1953); October 15, 1939, May 4, 1940, May 15, 1940, August 17, 1940, May 21, 1941, July 15, 1941, December 14, 1941, and May 31, 1944	AT	DE 717601; DE 718801; DE 718802; DE 721718; DE 736501 (DK 62838); DE 731607; DE 743146; DE 882085	Small centrifuge unit placed on the side of the tank, over the liquid surface
Kundt (1943); assigned to Hefe-Patent GmbH; June 19, 1941	DE	DE 743739	Foam breaker includes a collection surface, an impeller, and an extended round bottom
Stich (1944); October 29, 1941	DE	DE 744071	Aspirated with rotating sifters, and treated with jet pressures of diluted nutritive solution

(Continued)

Table 8—Continued.

Inventor (publication y); priority date	Country	Patent nr.	Details on foam treatment
Seidel (1956); January 23, 1943	DE	DE 943161	Foam is rapidly circulated and treated outside of the fermentation vessel
Seidel (1958); November 6, 1943	DE	DE 968601	Aeration system is designed to recirculate foam overflow which is returned to the bottom
Humfeld and others (1952); assigned to the Secretary of Agriculture, the United States; September 12, 1947	US	US 2610155	Foam breaker with an aeration device (patented separately, US 2542031)
Vogelbusch (1950); July 17, 1948	AT	AT 165869	Nutrients are sprinkled over the surface of the foam layer which is sucked to the bottom of the vat, and fed near the mixing device
Gordon and Veldhuis (1953); assigned to the Secretary of Agriculture, the United States; June 3, 1949	US	US 2635070	Broken by striking against a perforated plate placed adjacent to the outlet of a foam pipe; continuous culture
Traelnes (1969); assigned to Añico S.A. (Nestlé S.A.); April 13, 1967	CH	CH 465561 (GB 1169494; DE 1767168; FR 1558187; US 3520822)	A funnel-shaped chamber placed within the vessel, and equipped with a rotatable impeller
Vogelbusch GmbH (1970); Hofschneider (1971); March 14, 1969	AT	AT 290477 (FR 2037976; GB 1244577; DE 2010238; CH 505637)	Treated with a 2-stage centrifugal separator placed at the top of the vat
Müller (1975, 1976); assigned to Schering AG; September 6, 1973, and June 13, 1974	DE	DE 2345374 (GB 1485974); DE 2428312	Centrifugal foam destroyer into a small auxiliary vessel over the vat
Liepe and others (1976); November 12, 1975	DD	DD 122643	Removed, mechanically destroyed by intense shearing and torque, and centrifuged
Pilepp (1981); assigned to Fried. Krupp GmbH; February 12, 1980	DE	DE 3005115	Removed by aspiration, and recirculated through the aeration system

^a Breakthrough.

Especially in the early years of baker's yeast manufacturing, aeration produced so much foam that it could only be controlled by the addition of very large amounts of solid fat, such as lard, which had a negative effect on yeast keeping properties (Dampf-Kornbrennerei und Preszhefe-Fabriken AG 1909). According to Schmid and others (1986), "Now, in recent years, yeast manufacturers have complained that baker's yeast changes its consistency in storage, ultimately becoming soft or even liquid and hence unsalable. The reduced yeast quality is blamed by the yeast industry on the increasing surfactant content of the molasses. It has now been found that the consistency of baker's yeast is influenced by the nature of the foam inhibitor and that all the yeast defoamers hitherto available on the market or proposed for this purpose lead to more or less rapid softening of the yeast."

Many chemical antifoaming agents have been proposed in the early years of commercial baker's yeast, including such materials as long-chain liquid alcohols, vegetable phosphatides, and silicones (Steiner and others 1960). However, such preparations were only effective in large amounts so their use was uneconomical. Among other problems, fats and chemical antifoams were (1) expensive and impossible to recover, (2) made more difficult yeast separation and filtration from fermentation media, (3) contaminated yeast, and (4) affected negatively the yeast flavor. In addition, adding surfactants to yeast growth media reduced oxygen transfer and cell growth, which resulted in more intense aeration and mixing (Ghildyal and others 1988; Routledge and others 2014). For example, Henkel & Cie GmbH (1974) reported the following: "For de-foaming in the yeast industry, fatty oils such as rape oil, groundnut oil, or olive oil, as well as wool fat and mineral oil, have been used for a long time. Fatty acid monoglycerides, fatty acid polyglycol esters, polyalkyleneglycols, esters of tall oil acids, ethylene oxide adducts

of alkylphosphoric acids and branched fatty alcohols, alkylene oxide adducts of oligosaccharides, and free fatty alcohols have also been proposed for this purpose. Although in most cases a more or less strong suppression of the foam is attained with these substances, they could not be completely satisfactory for various reasons, particularly owing to the need for too large amounts. The high consumption is in some cases due to a not sufficiently long lasting action."

Safety of chemical antifoams is important because it is intended for human consumption. Substances used to regulate the foam must be largely removed in the usual working-up of the end-product, for instance in the washing of the yeast, so that they can no longer be detected in the finished product. Apart from this, all adjuvant substances which are used in the production of foodstuffs should be odorless and tasteless and, obviously, also physiologically completely harmless. For economic reasons, and in order to limit the contamination, highly active antifoaming agents are desirable. Foam is constantly formed during yeast growth, which requires the use of a device that will automatically inject antifoam when foam has attained a specific height. Such devices include floaters that may be lifted by the foam so this may provoke the injection of antifoam when in excess (Svenska Jästfabriks Aktiebolaget 1939).

There would be no universal antifoam. Efficient antifoams for fed-batch fermentation of baker's yeast on nitrogen-enriched molasses may not be appropriate for other fermentation media. A technique for testing chemical antifoams during yeast manufacture has been proposed by Morse and Moss (1952). Antifoaming agents must not be allowed to impair the growth of the yeast cells and, therefore, the yield of yeast. Onodera and others (1993) reported that the inhibitory effect of antifoam agents on yeast growth varied according to oxygen transfer rate in the fermentation vessel.

Table 9—Patents published between 1900 and 2014 on chemical antifoam for the production of baker's yeast.

Inventor (publication y); priority date	Country	Patent nr.	Details
Weissweiler (1922); February 7, 1919 ^a	AT	AT 87319	Homogenized emulsion of oil or fat
Steibelt (1934); assigned to I. G. Farbenindustrie AG; July 26, 1931 ^a	DE	DE 602087 (US 2085709)	Aliphatic alcohols, a mixture of hexadecanol, tetradecanol, and solvent; alternative to olive oil; the 1st chemical antifoam agent
Aktieselskabet Dansk Gaerings-Industri (1934); December 12, 1933	DK	DK 49622 (FR 776346; AT 146477)	Specific fats, including from coconut, cotton, and soya
Lighthipe (1944); assigned to Natl. Oil Products Co.; May 29, 1940	US	US 2346928	Partial ester of a polyhydric alcohol with a higher hydroxyl fatty acid blended with a hydrophobic substance
Brodersen and Quaadvlieg (1944, 1943); assigned to I. G. Farbenindustrie AG; September 17, 1941	DE	DE 743985; DE 744679	Paraffin derivatives
Fritz (1945); assigned to Natl. Oil Products Co.; September 22, 1942	US	US 2390212	Fatty esters of glycols and polyglycols
Wallerstein and Alba (1951); assigned to The Overly Biochemical Research Foundation, Inc.; September 28, 1944	US	US 2572823	Protein-containing solution (wheat and malt) treated with pancreatic protease
Goodwin (1950); assigned to Dow Corning Corp.; April 20, 1948	US	US 2511056	Organo siloxanes
Valko and Kamlet (1953); assigned to E. F. Drew & Co., Inc.; October 27, 1948	US	US 2635079	C-alkyl N-acylamido alkyl imidazolines
Walker and Farbak (1954); assigned to Swift & Co.; May 10, 1950	US	US 2668138	Ebony fat (residue from the refining of animal fat) mixed with fat from animal, vegetable or mineral source
Moss and Morse (1954); assigned to Monsanto Chemical Co.; July 14, 1950	US	US 2666017 (GB 700557)	Ethylene oxide derivatives from abietic acid and tall oil; see cited report by Morse and Moss (1952)
Kulakow (1956); assigned to Hodag Chemical Corp.; May 7, 1952	US	US 2762780	Sugar cane oil blended with an emulsifier
Hlavacek (1957); assigned to Swift & Co.; September 4, 1952	US	US 2809168	Sugar cane oil blended with a petroleum oil carrier
De Castro and others (1959); Grool and others (1955); assigned to Nopco Chemical Co.; February 12, 1953	US	US 2868734 (CA 513513)	Blend of (1) an aliphatic hydrocarbon, (2) a mixture of selected dihydric alcohol esters of higher fatty acids, and (3) a carboxylic acid type composition; forms very stable emulsions
Greve (1959, 1961); assigned to Schill & Seilacher; September 7, 1955, and February 11, 1956	DE	DE 1050289; DE 1102083	Polydimethylsiloxanes or polymethylsiloxanes
Steiner and others (1960); assigned to Swift & Co.; March 19, 1956	US	US 2931780	Sterol-free mixture of fatty acids, rosin acids, and unsaponifiable materials; mixture is derived from tall oil pitch
Pisiuk (1960); assigned to Fabriques de Levures, Sucre & Alcools de Saint-Ouen-L'Aumône, and Chiron; August 27, 1958	FR	FR 1210092 (LU 37580; BE 582023; GB 875113)	Organo polysiloxane
Baenitz and others (1968); assigned to Chemische Werke Witten GmbH; December 31, 1960	DE	DE 1263705 (FR 1308105; CH 408867; NL 127665; NL 269733)	Mixtures of partial esters prepared from specific fatty acids, for example, coconut oil
Heyden and Asbeck (1973); Henkel & Cie GmbH (1973); December 28, 1971	DE	DE 2164907 (FR 2166092; GB 1371731)	Ethylene oxide and propylene oxide; lasting antifoam effect
Heyden and Asbeck (1973); Henkel & Cie GmbH (1975); February 29, 1972	DE	DE 2209559 (GB1386042)	Mono- and/or diester of hydroxystearyl alcohol with a saturated fatty acid or hydroxyl-fatty acid having more than 16 carbons
Heyden and others (1977); assigned to Henkel & Cie GmbH; December 1, 1975	DE	DE 2553990 (US 4087398)	Higher (C ₂₂₋₇₀) aliphatic ketones and higher (C ₂₂₋₇₀) aliphatic secondary alcohols (solids with small particle size)
Heyden and others (1977); assigned to Henkel & Cie GmbH; December 22, 1975	DE	DE 2558040 (GB 1520539; US 4082689)	Bis- α -hydroxyamine derivative
Heyden and others (1977); assigned to Henkel & Cie GmbH; December 22, 1975	DE	DE 2557898 (GB 1521398; US 4094812)	α -hydroxyamine derivative
Schmid and others (1986); assigned to Henkel KGaA; September 24, 1984 (DE)	DE	EP 0180664 (US 4622303)	Polyglycol ethers containing from 1 to 3 ethylene glycol ether groups; to avoid yeast inhibition and premature softening
Dietsche and others (1986); assigned to BASF AG; October 19, 1984 (DE)	DE	EP 0180081 (US 4753885)	Fatty alcohol or alkylphenol oxyalkylates possessing blocked terminal groups
Lange and others (1987); assigned to Henkel KGaA; August 31, 1985 (DE)	DE	EP 0217116 (US 4745231)	Alkylene oxide block of polymers of glycerol and polyglycerol

(Continued)

Table 9—Continued.

Inventor (publication y); priority date	Country	Patent nr.	Details
Höfer and others (1987); assigned to Henkel KGaA; April 18, 1986	DE	DE 3613128 (US 4797222)	Block polymer mixed with an emulsifier (specific formula); linked to the above entry
Schmid (1989); assigned to Henkel KGaA; January 11, 1988 (DE)	DE	EP 0324340 (US 4942049)	Terminally blocked polyethoxylated fatty alcohols; odorless without yeast inhibition
Hill and others (1990); assigned to Henkel KGaA; May 5, 1989 (DE)	DE	EP 0396044 (US 5306442)	Mixture of alkyl glucosides and aliphatic alcohols; completely biodegradable
Hill and Mahler (1992); assigned to Henkel KGaA; July 27, 1990	DE	US 5154856 (WO 9202284)	Alkadienyl ethers of monosaccharides, disaccharides, and alkyl glucosides
Mahler and others (1992); assigned to Henkel KGaA; December 4, 1990	DE	DE 4038608 (WO 9210266; CA 2097829 and 7 others)	Ester polyols made by reacting, with ring opening, of epoxidized carboxylic acid esters with carboxylic acids and / or alcohols
Milan and others (1992); December 20, 1990	CS	CS 9006472 (CZ 277788)	Mixture of acyl glycerols and acyl polyglycerols and/or polyglycols
Mahler and others (1992); assigned to Henkel KGaA; December 24, 1990	DE	DE 4041754 (US 5378485)	Carbonic acid esters; no inhibitory activity against yeast
Härer and others (1992); assigned to Henkel KGaA; February 13, 1991	DE	DE 4104330	Phosphonyl silanes and/or phosphonyl silicones; high-lasting action without impact on yeast yield
Wangemann and others (1994); assigned to Henkel KGaA; July 30, 1992	DE	DE 4125236 (WO 9403251)	Terminal-group-closed antifoaming agents
Wollenweber and others (1994); assigned to Henkel KGaA; March 15, 1993 (DE)	DE	WO 9421348 (US 5725815)	Block copolymers containing alkylene oxide units
Möller and others (1995); assigned to Henkel KGaA; September 15, 1993 (DE)	DE	WO 9507742 (US 5710207 and 7 others)	Ester group-containing block polymer, containing ethylene glycol ether units esterified with optionally polymeric hydroxycarboxylic acids
Schmid and Bigorra Llosas (1996); assigned to Henkel KGaA; September 1, 1994 (DE)	DE	WO 9606905 (US 5811594)	Methyl-end-capped alkyl and /or alkenyl polyglycerol ethers
Wollenweber and others (1996); assigned to Henkel KGaA; February 1, 1995 (DE)	DE	WO 9623568 (US 6057375)	Alkoxylation products of epoxidized fatty compounds
Wollenweber and others (1996); assigned to Henkel KGaA; February 16, 1995 (DE)	DE	WO 9625215	Aqueous dispersions of alcohols, the oil phase of which is solid at room temperature and also with optionally esterified glycerine polyglycol ethers or polyglycerine ethers which are liquid at room temperature
Oftring and others (1997); assigned to BASF AG; July 20, 1995	DE	DE 19526501	Mixed hydroxyethers
Wollenweber and others (1998); assigned to Henkel KGaA; July 19, 1996 (DE)	DE	WO 9803243	Ethylene oxide—propylene oxide addition compounds of glycerol or polyglycerol reacted with alpha-olefin epoxides
Wollenweber and others (1998); assigned to Henkel KGaA; January 27, 1997	DE	DE 19702846	Includes an alkyl-glycoside nonionic dispersant, and a reaction product with mono- or polyhydric alcohol(s)
Löhl and others (2000); Tundo and others (2000); assigned to Cognis Deutschland GmbH; May 5, 1999	DE	DE 19920559 (EP 1050524)	Alkyl and/or alkenyl ethers with terminal alkyl groups
Browne (2004); assigned to Bayer Polymers LLC; November 7, 2002 (US)	US	US 7001634 (WO 2004043567 and 11 others)	Contains hydroxyl equivalent molecular weight of at least 4000 Da
Wurm and Marques (2010); Assigned to The Dow Chemical Co.; September 24, 2008 (US)	US	US 20100075389 (WO 20100036555; EP 2346925)	Alkylene oxide capped secondary alcohol ethoxylates; mainly for bioethanol production

^a Breakthrough.

Major Breakthroughs

A few important innovations have been disclosed in patents on aeration and foam control in baker's yeast production.

A. Aeration control. Aerating yeast growth media was a breakthrough in the 19th century. It was not until 1928 that Norddeutsche Hefeindustrie AG proposed to increase progressively the aeration rate in the growth medium, to reduce energy costs. In 1938, fine-tuning of aeration was also patented by Rosenqvist from a Swedish company, Svenska Jästfabriks Aktiebolaget. Continuous measurement of oxygen was specifically proposed around 1959 to 1963 by the same company and also by Hefe-Patent GmbH.

B. Air circulation and distribution. Many inventors proposed innovative solutions as the best way to inject air into fermentation tanks. A lasting solution was to introduce air at the bottom of the tank. For example, in 1901 Klipfel proposed to introduce air through a series of perforated tubes. In 1927, Stich recommended a bottom air distribution system equipped with pores to give very fine air bubbles and improve oxygen transfer. Around 1930 to 1934, several technological solutions were developed to further improve oxygen transfer, for example, by Vogelbusch in 1932. Since this period, several innovative solutions were proposed to further improve oxygen transfer and reduce aeration, although it is difficult to determine which

Table 10—Patents published between 1900 and 2014 on the distribution of chemical antifoam during the production of baker's yeast.

Inventor (publication y); priority date	Country	Patent nr.	Details
Schumacher (1913); August 30, 1912	AT-HU (AT)	AT 258201	Oil addition is controlled by a vibrating shell which liquefies the foam, and pushes down the bowl
von Hoesslin (1920a, 1920b); November 14, 1916, and April 14, 1917	DE	DE 317918; DE 318894	Addition of fat (oil) emulsified with water vapor is controlled by a pump connected to air compressor
Ostwerke AG, Zweigniederlassung Wandsbek-Hamburg (1927); February 21, 1925 ^a	DE	DE 448846	Valve for fat addition is opened by an electrical device controlled by a float
Boye (1927); July 4, 1925	SE	DE 469117 (FR 616277; US 1792963)	Spray of heated fat is controlled by a float; to avoid hand addition of fat and misjudgement by operators
The Distillers Co. Ltd. and Meyer (1931); May 16, 1930	GB	GB 349288	Foam enters a container, and ejects antifoam agent
Harrison (1933); assigned to Standard Brands Inc.; July 2, 1930	US	US 1916614	Oil addition is controlled with a float
The Distillers Co. Ltd., and Everett (1933); licensed to Standard Brands Inc. (US only); June 10, 1932	GB	GB 402826 (US 1984789)	Photoelectric cell; to reduce the amount of fat added, and control infection
The Distillers Co. Ltd., and Montgomery (1933); November 21, 1932	GB	GB 397722	Distributor is responsive to the air pressure created by rising froth
Ries (1937); April 29, 1936	DE	DE 648672	Electromagnetic distributor
Svenska Jästfabriks Aktiebolaget (1939); Andrén (1943); December 22, 1937 (SE)	SE	FR 847926 (DE 730846)	Distributor adds a fixed quantity of fat, using a float
Aktieselskabet de Danske Spritfabrikker (1941); February 2, 1940	DK	DK 58982 (GB 572759)	Regulated by an electrical circuit produced between an electrode placed above the liquid and the liquid itself
Broch (1948); assigned to État Français (Ministre de l'Armement); January 24, 1947	FR	FR 940515	Antifoam is added through the aeration system and in a finely divided state (aerosol)
Lehmann (1960); assigned to Schill & Seilacher; October 25, 1958	DE	DE 1076628	Foam is blown outside of the main vat where antifoam is added; alternative to direct injection on foam
Patentauswertung Vogelbusch GmbH (1962a, 1962b); Rungaldier and Braun (1966, 1967); June 20, 1960, and April 10, 1961	AT	AT 219537; AT 223569 (DE 1209089; GB 951674; US 3249515; US 3354050; FR 1290557)	Foam is passed through a separator, and antifoam agent is added in amount dependent upon rate at which the liquid is separated; less antifoam agent
Richards and others (1971); assigned to The Distillers Co. (Yeast) Ltd.; June 20, 1968	GB	GB 1236817 (DE 1931077; LU 58904)	A pneumatic control system delivers a predetermined quantity
Li (2012); assigned to Angel Yeast Co. Ltd.; December 8, 2011	CN	CN 202337788	Low-cost defoamer dosing device equipped with a probe-type level transmitter

^a Breakthrough.

innovative solutions truly resulted in the highest energy savings and foam reduction. For example, multiple attempts have been made to aerate growth media outside of the main fermentation vat.

- C. Foam control. The 1st patented invention that addressed the removal of excess foam from yeast media was filed in 1900 by Raben and Wrede who proposed a mechanical device. More success was obtained with chemical defoamers, especially specific fat compositions. The 1st patent on chemical antifoams for baker's yeast was filed in Austria in 1919 by Weissweiler who proposed to use a fat emulsion. In 1931, Steibelt from Farbenindustrie AG designed the 1st proprietary composition of antifoam, based on aliphatic alcohols. Among the many antifoam compositions that were patented thereafter, not less than 22 formulations were proposed between 1971 and 1997 by Henkel & Cie GmbH.

Technological Lessons from Patents on Aeration and Foam Control

The development of innovative aeration systems and foam control devices was critical to the modern baker's yeast industry, with 245 patented inventions. This number is about the same as on

fermentation control, although more interest was given to growth media with about 350 original patent specifications (Gélinas 2012, 2014). Compared to media development, control of aeration and foam probably required more effort by inventors because specific equipment was designed and it generally involved much engineering input. With specific aeration devices, the baker's yeast industry would go beyond artisan-like formulations of growth media and fermentation control. Although it started as early as 1879, interest for inventions on aeration and foam control remained active for more than 100 y or until the end of the 20th century. While the nature of molasses-based media was highly specific, technological solutions in the yeast industry probably helped other fermentation industries with aeration, including antibiotics manufacture. For example, commercial citric acid production from aerobic fermentation was introduced in the 1950s that is about 30 to 40 y after modern commercial baker's manufacturing (Abou-Zeid and Ashley 1984).

Baker's yeast strains have been progressively selected as a result of intensive domestication under high aeration combined with carbon-limited conditions (Gélinas 2009; Randez-Gil and others 2013). Domesticated yeast cells now require very large quantities of oxygen so it is not possible to find, in nature, yeast cells which have been grown under such conditions.

Impact on the Baking Industry

Intense aeration was the single-most critical parameter to markedly increase the yield of yeast cell biomass from a specific weight of molasses. For bakers, this greatly contributed to cost reduction of this important ingredient. Aerated yeast has improved keeping properties because strong aerobic conditions help yeast cells to store carbohydrates such as trehalose and glycogen (Gélinas and others 1989). On the contrary, so-called alcohol yeast did not keep well and was not suitable for baking applications, except when in a very fresh state.

According to many inventors, growth under intense aerobic conditions markedly reduced yeast gassing power by strongly inhibiting the ability to ferment maltose, and this was later confirmed by scientific studies (Suomalainen and Oura 1956; Beran and others 1963; Suomalainen 1963). In the 1st steps of yeast production, cells are grown under anaerobic conditions, then switched to strong aerobiosis, and suddenly brought back to anaerobiosis when mixed into the bread dough (Suomalainen 1963). In essence, when aerobic conditions became the norm in the industry, baker's yeast strains were not well adapted to a maltose-rich environment such as nonsugared dough.

In addition to aeration, growth media were also very important. Prior to the advent of air yeast, maltose-rich cereal-based media were used in the early days of baker's yeast manufacturing before being replaced by maltose-poor molasses (Gélinas 2012). Grain yeast was richer in maltase and its gassing power in nonsugared bread dough was faster compared to yeast prepared exclusively from molasses, because cells had to adapt to maltose, their main sugar source (Tipper 1954). Small addition of sucrose was a very efficient procedure to compensate for variations in baker's yeast maltase activity (White 1954). Over the years, maltose fermentation ability of baker's yeast in bread dough was improved through strain selection and cultivation conditions including specific proportions of beet and cane molasses (Pyke and others 1958; Beran and others 1963). Later, Khorola and others (1982) did not report reduced gassing power in yeast with decreased maltose fermentation, which was attributed to high levels of α -glucosidase. In the 1980s, patented strains were developed for unsugared dough, and so-called lagging strains also had high gassing power in high-sugar dough (Gélinas 2009).

The use of baker's yeast showing reduced maltase activity results in variable delays in dough fermentation in the 2nd hour of leavening, which has later led to several studies on differences in genetic background of the strains (Higgins and others 1999). It is likely that, in the early days of industrial bread making and yeast manufacturing, bakers gradually considered sucrose as an essential ingredient to stimulate yeast gassing power, especially where high-volume pan bread was popular such as in England and the United States. In the early days of commercial baker's yeast, bread formulations in Great Britain and many other countries contained very little sugar. Essentially considered as a yeast food, sugar was used at low concentrations in dough, similar to yeast levels (Simmons 1903). In the United States, the level of sweeteners in pan bread increased from about 2% in 1920 to 8% in 1958 (Bohn and Junk 1960), which parallels the development of the baker's yeast industry, including the widespread use of efficient aeration techniques. For example, sugar levels as high as 12% on a flour basis have been reported in hamburger buns (Kulp and Ponte 2000). Addition of high levels of sugar in dough has been shown to result in bread enrichment with poorly digestible fructose (Gélinas and others 2016).

Hence, the advent of aerated yeast was accompanied by variable gassing activity in dough which, among other factors, probably contributed to the popularity of bread sweetening; such practice considerably changed the taste and nutritive value of bread. In the United States, complex business obstacles, costs, and consumer confusion would now be associated with labeling added sugars in food, including bread (Goldfein and Slavin 2015).

To a lesser extent, the inevitable addition of very large quantities of surfactants in yeast growth media to control excess foaming also changed the composition of yeast cells and had a negative effect on its keeping properties. Hence, it is likely that those surfactants remain associated with yeast, so most bread is likely to contain traces of antifoam, a situation that was not found prior to the development of aerated yeast around 1900 to 1915.

Conclusions

Intense aeration during growth completely changed baker's yeast properties. Domesticated yeast cells are now addicted to high levels of oxygen. It is not possible to find similar yeast cells in nature. Inventors took the matter so seriously that not less than 245 patent specifications were filed to improve the aeration of fermentation vats with the inevitable need to control excess foam. This has led to the development of specialized industries that had to adopt concepts such as oxygen transfer and the specific nature of foam produced from nitrogen-enriched molasses. Most of these technologies were developed in the early 1930s but more efficient aeration conditions and, especially, foam control in baker's yeast production are still being searched. Indirectly, the widespread use of intense aeration for growing baker's yeast impaired its fermentative activity in anaerobic bread dough which encouraged bakers to supplement the latter with sugar.

Conflict of interest

There is no conflict of interest.

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